

MAT 310, Handout 17: MGFs, Tchebysheff's Theorem, and Mixed Distributions (4.9-4.11)

Moment-Generating Function (mgf) of a Continuous RV, Y :

$$m(t) =$$

Example 1. Find the mgf of $Y \sim U(a, b)$.

Example 2. Find the mgf of $Y \sim \exp(\beta)$.

Tchebysheff's Theorem: Let Y be a RV with mean μ and variance σ^2 . Then, for any constant $k > 0$,

$$P(|Y - \mu| < k\sigma) \geq$$

Equivalently,

$$P(|Y - \mu| \geq k\sigma) \leq$$

Proof (of the continuous case):

Remark: Note that the same Tchebysheff inequality statements apply to both discrete and continuous RVs.

Example 3. A wind turbine blade has an expected lifespan of 25 years with a standard deviation of 3.5 years. Find an interval that contains the lifespan of a turbine blade with probability .80.

Mixed Distribution: Let Y be a RV that follows the distribution of Y_1 w.p. c_1 , Y_2 w.p. c_2 , ..., and Y_n w.p. c_n , such that $\sum_{i=1}^n c_i = 1$. In other words, c_i = probability that any random observation is from distribution i . Then,

$$F(y) = c_1 F_1(y) + c_2 F_2(y) + \dots + c_n F_n(y)$$

$$E(Y) = c_1 E(Y_1) + c_2 E(Y_2) + \dots + c_n E(Y_n)$$

$$E[g(Y)] = c_1 E[g(Y_1)] + c_2 E[g(Y_2)] + \dots + c_n E[g(Y_n)]$$

Example 4. Suppose that on a randomly selected day it does not rain with probability 2/3. If it does rain, the amount of rainfall is exponentially distributed with a mean of 2 cm.

(a) Find the daily expected rainfall.

(b) Find the probability that it rains more than 2 cm on any given day.

(c) Sketch $F(y)$.